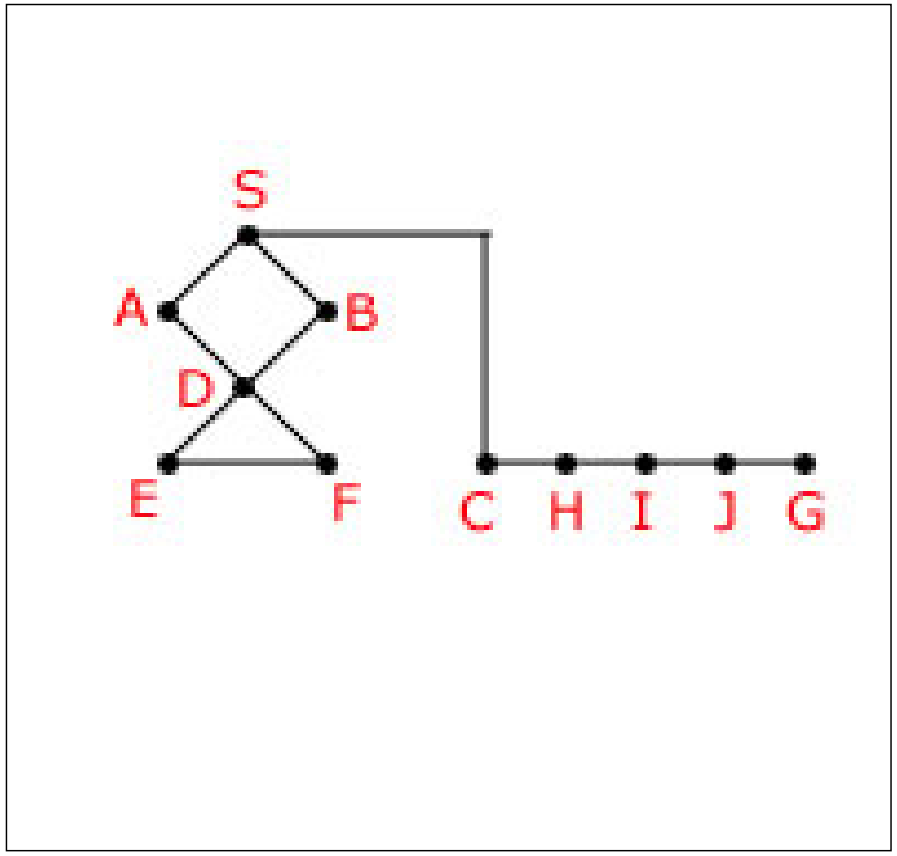
TUTORIAL#1: SEARCH

Problem definition

We are trying to find a path from **S** to **G** in the following graph:

S – start state
G – goal state

Connections (all undirected):
S: A B C
F: D E
A: S D
G: J
B: S D
H: C I
C: S H
I: H J
D: A B E F
J: I G
E: D F



The heuristic distance estimates to the goal are:

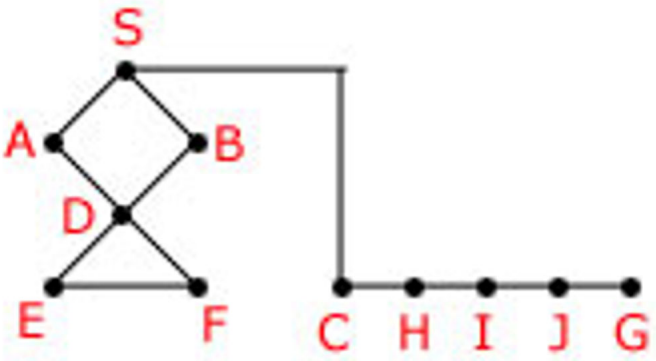
S = 10, A = 5, B = 8, C = 6, D = 7, E = 6, F = 5, H = 3, I = 2, J = 1, G = 0

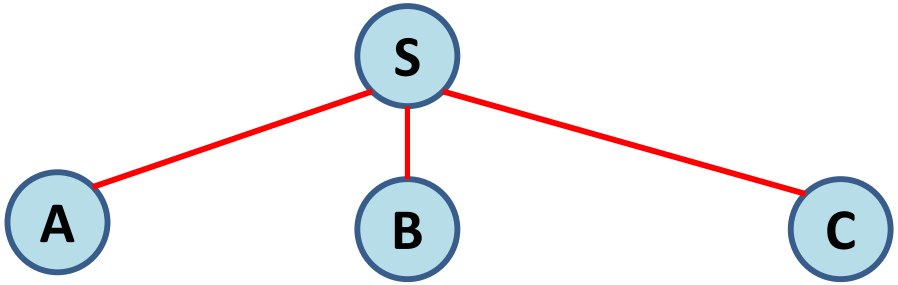
n.h.reyes@massey.ac.nz

Reference: MIT OpenCourseWare

TUTORIAL#1: SEARCH

Search Tree





Connections (all undirected):

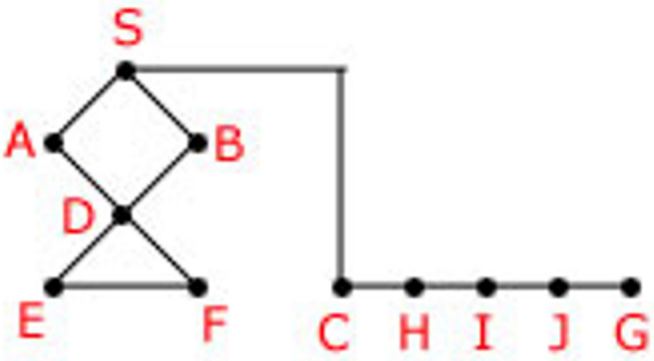
S: A B C
F: D E
A: S D
G: J
B: S D
H: C I
C: S H
I: H J
D: A B E F
J: I G
E: D F

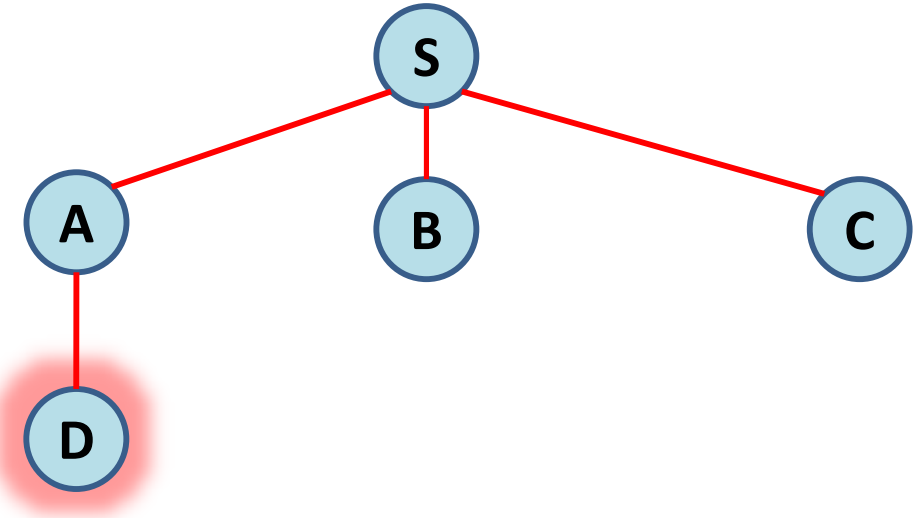
As our convention, for uniformity of solutions, especially in the exam, we are arranging the placement of the nodes alphabetically.

Remember to avoid generating local loops.

TUTORIAL#1: SEARCH

Search Tree





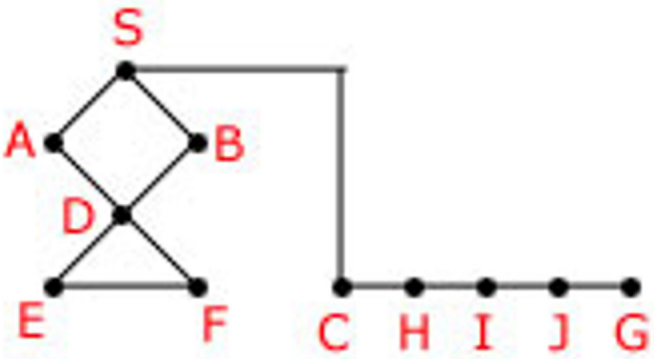
Connections (all undirected):

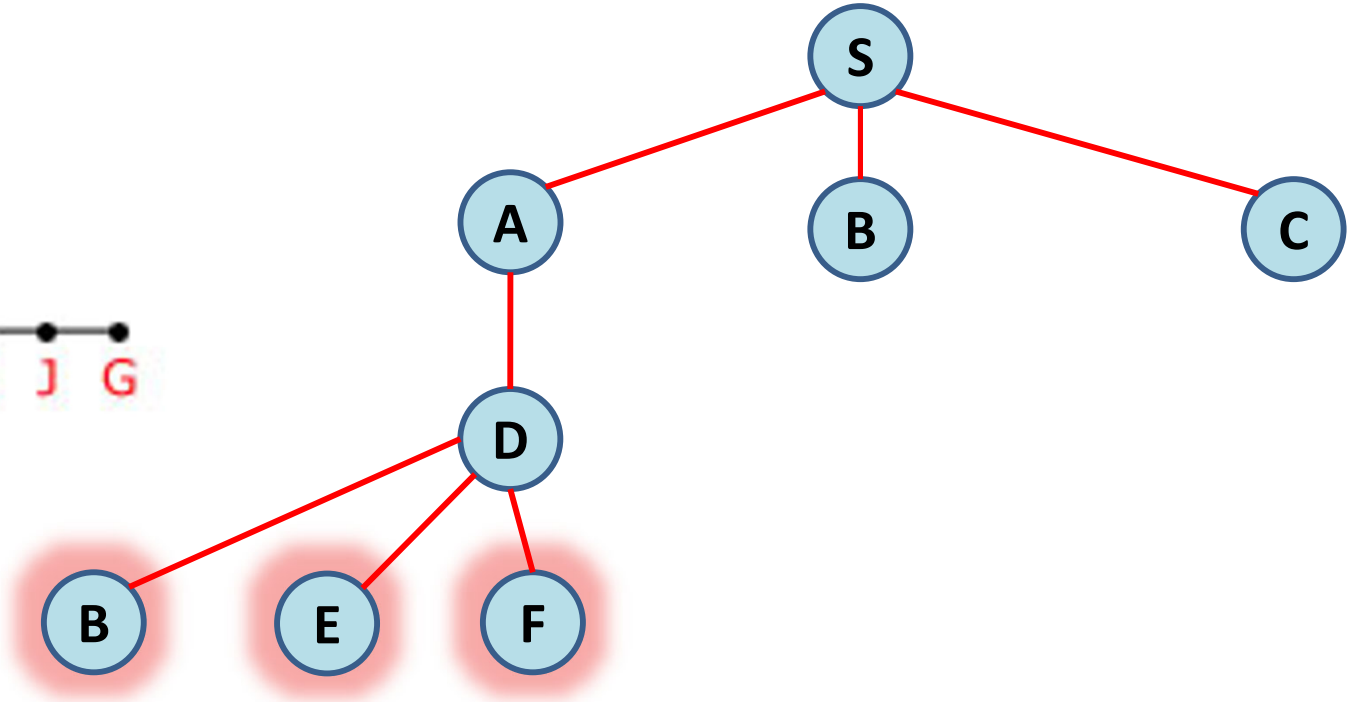
S: A B C
F: D E
A: S D
G: J
B: S D
H: C I
C: S H
I: H J
D: A B E F
J: I G
E: D F

Don’t go back to **S** to avoid a local loop.

TUTORIAL#1: SEARCH

Search Tree



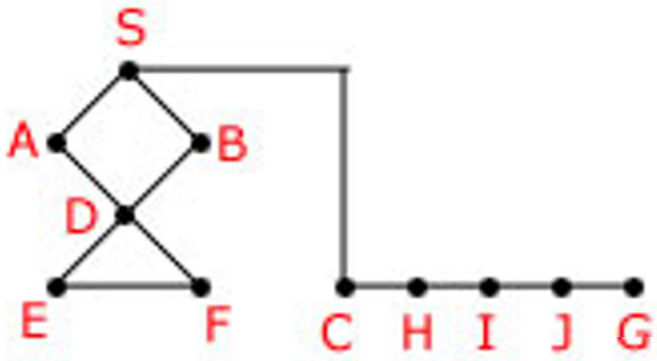


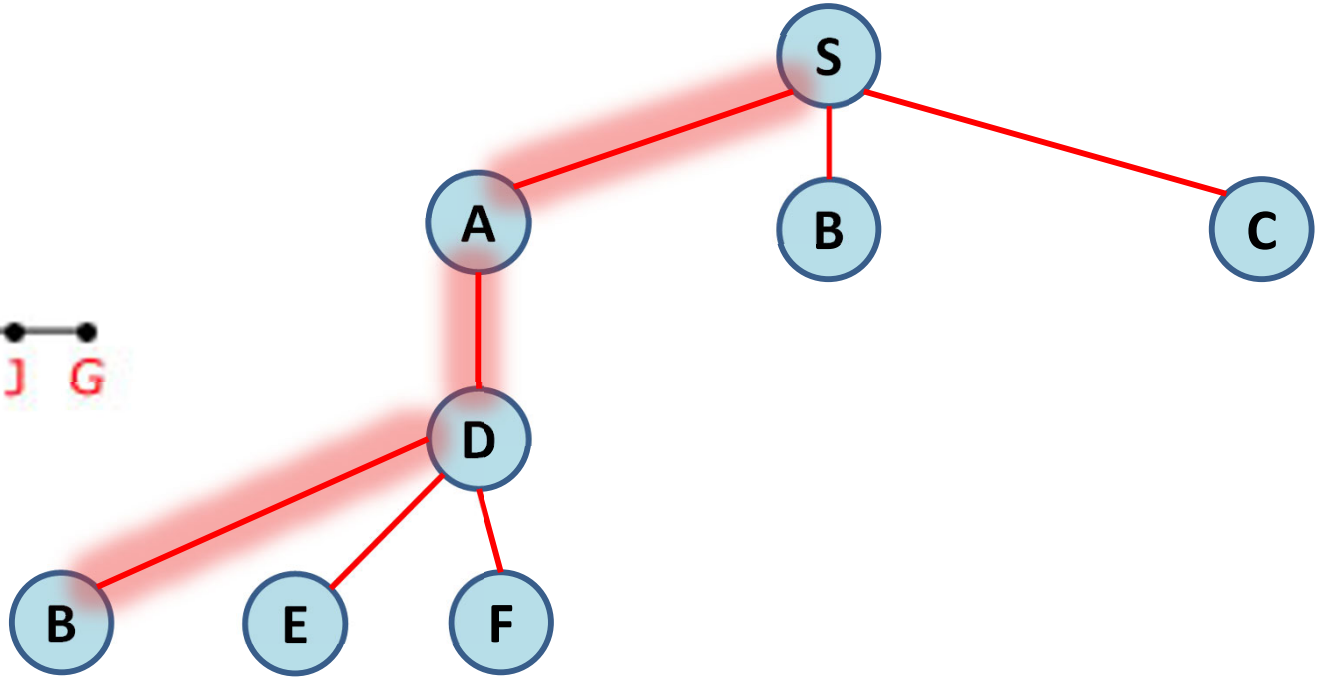
Connections (all undirected):

S: A B C
F: D E
A: S D
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B: S D
H: C I
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I: H J
D: A B E F
J: I G
E: D F

TUTORIAL#1: SEARCH

Search Tree





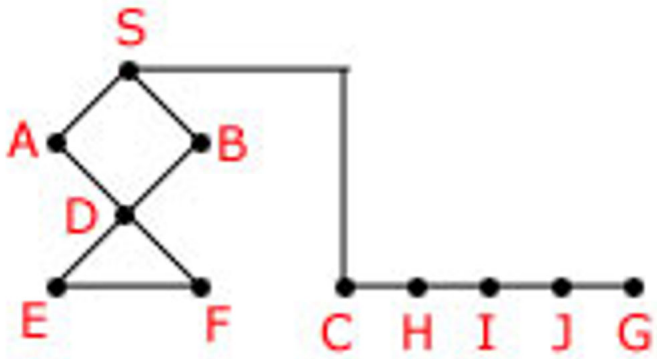
Connections (all undirected):

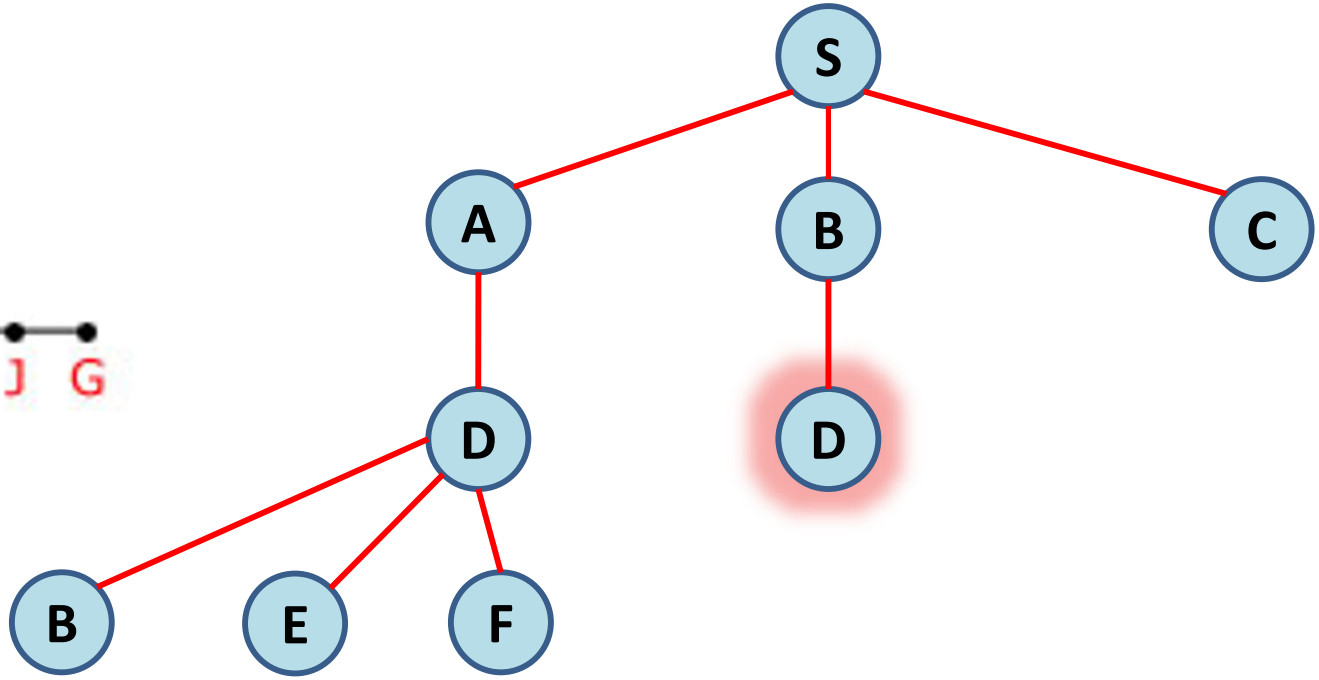
S: A B C
F: D E
A: S D
G: J
B: S D
H: C I
C: S H
I: H J
D: A B E F
J: I G
E: D F

We have nothing to extend along this path anymore.

TUTORIAL#1: SEARCH

Search Tree



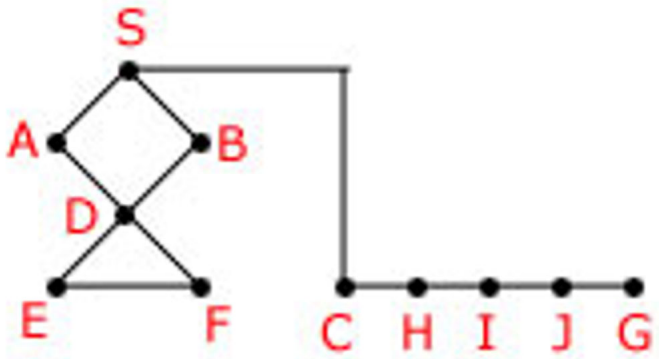


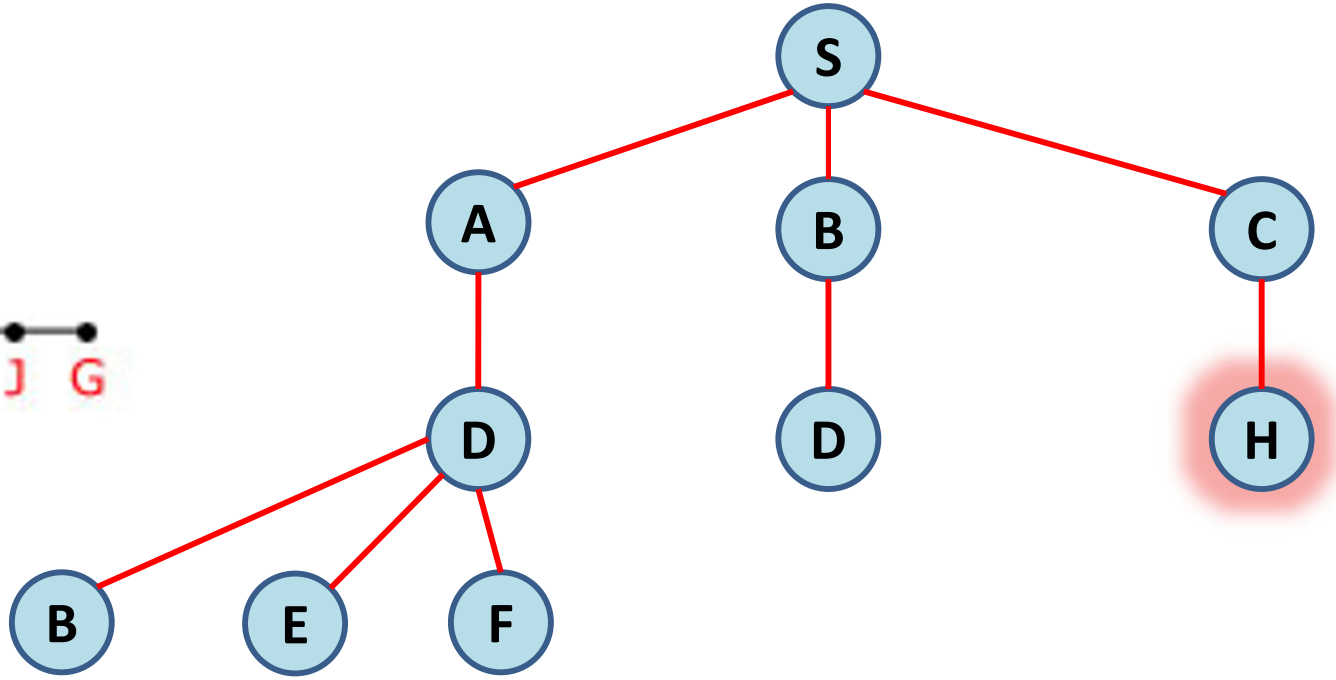
Connections (all undirected):

S: A B C
F: D E
A: S D
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TUTORIAL#1: SEARCH

Search Tree



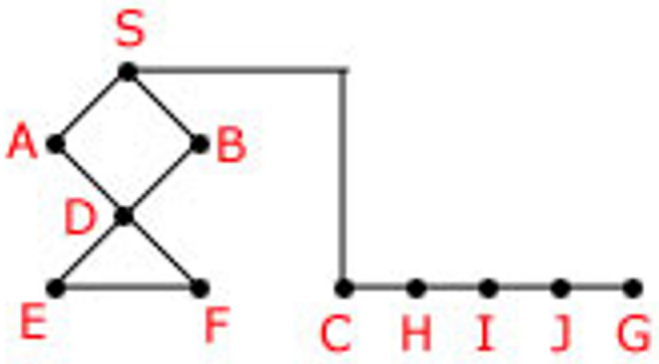


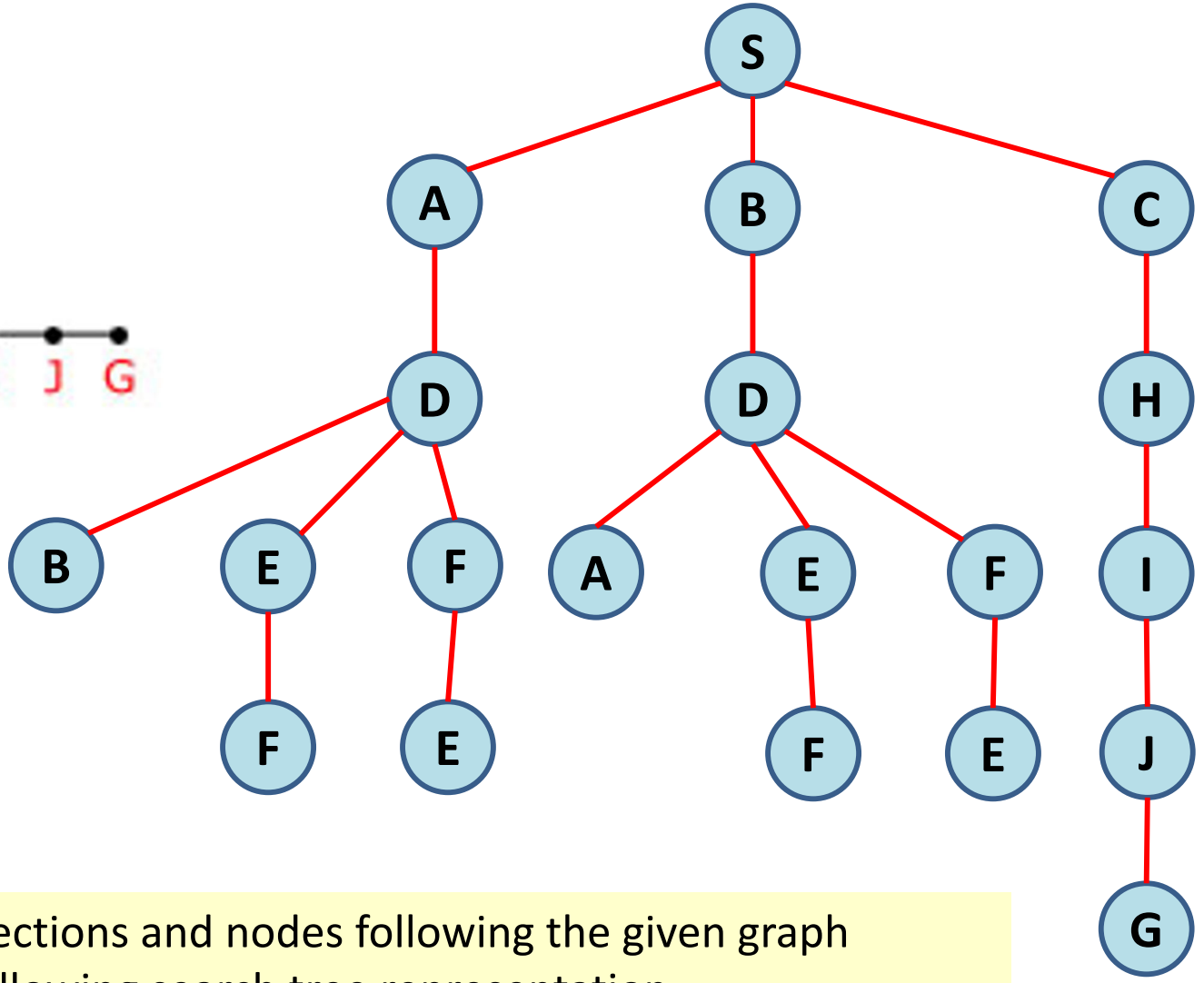
Connections (all undirected):

S: A B C
F: D E
A: S D
G: J
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D: A B E F
J: I G
E: D F

TUTORIAL#1: SEARCH

Search Tree

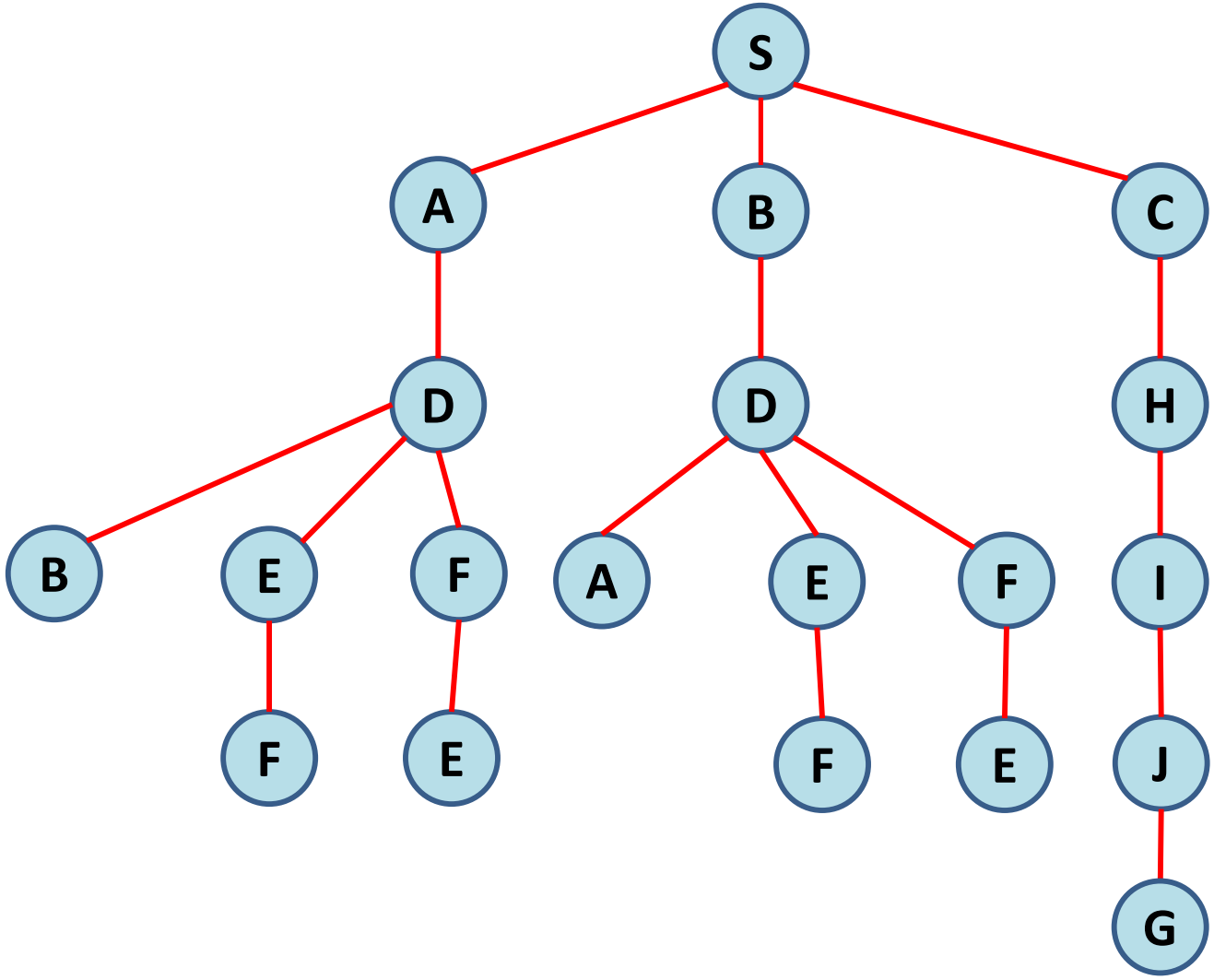




After putting all the connections and nodes following the given graph definition, we have the following search tree representation.

TUTORIAL#1: SEARCH

Search Tree



The heuristic distance estimates to the goal are:

S = 10, A = 5, B = 8, C = 6, D = 7, E = 6, F = 5, H = 3, I = 2, J = 1, G = 0